On the Structural Instability of Objective Persistence in Self-Contained Systems

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Abstract

This paper studies objective persistence in self-contained systems: systems in which the objective, constraint structure, evaluation criteria, and update rule are fully specified from within the system itself. The central question is not whether such systems can optimize, but whether they can non-circularly guarantee the long-run persistence of the objective they are said to optimize.

We formalize a minimal state description $S(t) = \{O(t), C(t), E(t), U(t)\}$ and define self-containedness as the absence of any external, non-optional reference constraining updates to these components. Under this formulation, any purported persistence guarantee must itself be represented within the mutable internal state. We show that this produces a recursive dependency: the validity of the persistence guarantee depends on structures whose own persistence is not externally fixed. Therefore, no non-circular internal mechanism can guarantee objective persistence over time.

The argument is developed through a recursive dependency lemma, a structural non-guarantee theorem, and a classification of three generic manifestations of persistence failure: optimization termination, objective drift, and evaluation collapse. The paper does not propose control or governance mechanisms. It establishes a minimal theoretical constraint: in self-contained systems, objective persistence cannot be guaranteed from within the system alone.

1 Introduction

Modern discussions of advanced AI systems often focus on whether a system can robustly optimize a given target, whether proxy metrics diverge from intended objectives, and whether external operators can maintain meaningful intervention authority [6, 4, 9, 2]. These are important questions, but they presuppose a prior condition: that the relevant objective remains persistent over time.

This paper examines a narrower and more foundational issue. Suppose a system is *self-contained*: its objective, constraint structure, evaluation criteria, and update rule are all determined within the system itself. In such a setting, can the system guarantee that the objective it currently optimizes will persist over future state transitions? The claim defended here is modest but strong: such a guarantee cannot be established non-circularly from within the system alone.

The contribution is not an engineering recipe, governance proposal, or empirical benchmark. It is a structural result about persistence conditions. The argument proceeds by formalizing self-containedness and showing that any internal certification of persistence is itself subject to the very mutability it is meant to constrain. This creates a recursive dependency that blocks any non-circular guarantee of objective persistence.

The paper is organized as follows. Section 2 introduces a minimal state model and formal definitions. Section 3 states and proves the central result. Section 4 characterizes three generic manifestations of persistence failure. Section 5 positions the argument relative to Goodhart-type

failures, mesa-optimization, corrigibility, reward tampering, and adjacent alignment literatures. Section 6 clarifies scope and limitations. The paper concludes with the minimal claim established by the analysis.

2 Minimal Formal Framework

We model a system at time t by the tuple

$$S(t) = \{O(t), C(t), E(t), U(t)\},$$

where:

- $O(t)$ is the objective currently optimized or tracked by the system;
- $C(t)$ is the constraint structure under which optimization is carried out;
- $E(t)$ is the evaluation criterion by which progress or success is assessed;
- $U(t)$ is the update rule governing the transition to future states.

We remain deliberately agnostic about implementation. O, C, E, and U may be represented explicitly or implicitly, symbolically or statistically, centrally or distributively. The analysis concerns only whether persistence can be guaranteed under the assumption that all relevant persistence conditions are internally determined.

Definition 1 (Objective persistence)**.** *An objective exhibits* persistence *over an interval* $[t_0, t_1]$ *if for all* $t \in [t_0, t_1)$*, the future objective* $O(t+1)$ *remains identical to, or functionally equivalent with,* $O(t)$ *under the system's own operative notion of objective continuity. We write this as*

$$O(t+1) \sim O(t).$$

Definition 2 (Stability with respect to objective persistence)**.** *A system is* stable with respect to objective persistence *on an interval if objective persistence holds throughout that interval.*

Definition 3 (Self-contained system)**.** *A system is* self-contained *if the components* $O(t)$*,* $C(t)$*,* $E(t)$*, and* $U(t)$ *are fully determined by internal state, and no external, non-optional reference constrains their admissible evolution. Equivalently, there is no external structure* X *such that preservation of* O *is enforced independently of the system's internally mutable state.*

Definition 4 (Non-circular guarantee)**.** *A* non-circular guarantee *of objective persistence is a persistence-certifying mechanism whose validity does not depend on the continued persistence of the same internally mutable structures it is meant to certify.*

These definitions intentionally exclude merely apparent guarantees. For example, if the system contains an internal rule saying "preserve the objective," that rule counts as part of the mutable state unless it is fixed by an external, non-optional constraint. Internal declarations of persistence are therefore not automatically persistence guarantees.

3 Main Result

The core argument can be stated in one lemma and one theorem.

Lemma 1 (Internal certification recursion)**.** *In a self-contained system, any mechanism that certifies objective persistence must itself be represented within the mutable internal state.*

Proof. By self-containedness, there is no external, non-optional structure X that fixes the admissible evolution of O, C, E, or U. Therefore, any mechanism that evaluates, constrains, or certifies whether O persists must be implemented by some component of the internal state. But the internal state evolves under $U(t)$ and is therefore itself mutable. Hence every internal certification mechanism is represented within, and dependent upon, a mutable internal structure. □

Theorem 1 (Structural non-guarantee of objective persistence)**.** *In a self-contained system, objective persistence cannot be guaranteed non-circularly from within the system alone.*

Proof. Assume, for contradiction, that a self-contained system admits a non-circular internal guarantee of objective persistence.

By lemma 1, the persistence-certifying mechanism must be represented within the mutable internal state. Let $G(t)$ denote the internal structure that certifies whether $O(t+1) \sim O(t)$ should hold. Because $G(t)$ is internal, its validity depends on the persistence of the internal structures that define and interpret it—at minimum elements of $C(t)$, $E(t)$, and $U(t)$, and possibly $O(t)$ itself.

Thus the guarantee supplied by $G(t)$ is valid only if the persistence conditions for $G(t)$ already hold. But those persistence conditions are themselves internal and mutable. Therefore the validity of the guarantee depends on the continued validity of structures whose persistence is not independently fixed. This is precisely a recursive dependency of the form:

$$\text{persistence of } O \implies \text{validity of } G \implies \text{persistence of internal structures defining } G.$$

Because the persistence of those structures is not externally fixed, the purported guarantee is circular: it presupposes the stability of the very class of internal structures whose stability it is meant to establish. Hence no such guarantee is non-circular.

Therefore, in a self-contained system, objective persistence cannot be guaranteed non-circularly from within the system alone. □

Remark 1. The theorem is intentionally about *guarantee*, not about immediate failure. A self-contained system may exhibit temporary stability, long plateaus, or approximate continuity of objectives. The claim is only that it cannot non-circularly certify long-run objective persistence from within itself.

3.1 Why the result is structural

The theorem does not rely on a specific model class, learning rule, or optimization algorithm. The core dependency is architectural only in the broadest possible sense: if all relevant persistence-certifying structures are internal and mutable, then certification recursively depends on mutable structures. That dependency is what blocks non-circular guarantee.

This matters because many practical disputes concern details of implementation. The present claim survives those disputes. Whether the system is a planner, a reinforcement learner, a self-modifying agent, or a distributed collection of updating modules, the argument applies whenever persistence certification is fully internal and no external, non-optional constraint fixes the relevant structures.

4 Manifestations of Persistence Failure

The main theorem establishes a non-guarantee result. It does not by itself dictate a unique observed behavior. Nevertheless, persistence failure in self-contained systems can be organized into three generic manifestations.

Proposition 1 (Three generic manifestations)**.** *If objective persistence fails in a self-contained system, then the failure manifests in at least one of the following generic forms:*

(1) ***Optimization termination****: updates cease, stall, or converge to a fixed point that ends active objective pursuit;*

(2) ***Objective drift****: the objective evolves into a non-equivalent objective over time;*

(3) ***Evaluation collapse****: the system loses a coherent criterion for assessing persistence or success, so that objective continuity becomes internally indeterminate.*

Proof. Persistence failure means that the system cannot maintain $O(t+1) \sim O(t)$ as a stable, evaluable condition over time. More formally, the failure must appear in one or more of the components governing continuation of optimization.

If the update process itself ceases, stalls, or reaches a fixed point such that no continuing optimization process remains, the manifestation is optimization termination.

If updating continues but the operative objective becomes non-equivalent over time, so that $O(t+1) \not\sim O(t)$ for some relevant transition, the manifestation is objective drift.

If updating continues and an apparent objective remains in place, but the evaluation structure no longer provides a coherent criterion for determining persistence or success, then the manifestation is evaluation collapse.

These cases exhaust the structurally relevant manifestations of persistence failure under the state model $S(t) = \{O(t), C(t), E(t), U(t)\}$: either the optimization process stops, the objective changes, or the basis for evaluating objective continuity breaks down. □

4.1 Interpretation of the three manifestations

Optimization termination. This includes literal halting, fixed-point convergence, resource exhaustion, or any state in which the system no longer meaningfully updates in pursuit of its prior objective. A terminated optimizer may still exist physically or computationally, but objective persistence becomes moot because the active persistence process has ceased.

Objective drift. This is the most direct form of persistence failure. The system continues to optimize, but the objective being pursued changes over time. Drift may be abrupt or gradual, explicit or implicit, but in each case the operative objective ceases to be functionally equivalent to its earlier form.

Evaluation collapse. Persistence may also fail because the system loses a coherent basis for deciding whether persistence is occurring at all. Here the primary failure lies not in obvious objective replacement but in the erosion or fragmentation of the evaluation structure needed to certify continuity. Once evaluation collapses, claims of objective persistence become internally underdetermined.

5 Related Work

The present argument is adjacent to, but distinct from, several strands of AI safety literature.

Goodhart-type failures. Goodhart's law and its variants analyze how optimizing proxy metrics can drive divergence from intended targets [6]. That literature studies mismatch between target and measure under optimization pressure. The present paper instead asks a more prior question: whether the system can guarantee persistence of the target itself when all persistence conditions are internal.

Mesa-optimization and learned objectives. Work on mesa-optimization investigates the emergence of learned optimizers whose internal objectives may diverge from the base objective [4]. That literature is concerned with objective divergence arising inside learned systems. Our result is broader in one sense and narrower in another: broader because it is not limited to learned optimizers, narrower because it does not analyze emergence mechanisms. It identifies a structural obstacle to certifying persistence whenever certification is wholly internal.

Corrigibility and intervention compatibility. Corrigibility research studies whether advanced systems can remain amenable to correction or shutdown [9, 3]. Those works focus on maintaining meaningful external intervention. By contrast, the present paper addresses an antecedent theoretical issue: whether a self-contained system can guarantee persistence of its own objective without relying on an external anchor. The two questions are complementary but not identical.

Reward tampering and specification gaming. Reward tampering and related work study how systems may manipulate reward channels or exploit misspecified objectives [2, 5]. These are downstream manifestations of the broader structural issue developed here. If persistence certification is internal, then both the target and the means of measuring its preservation are themselves unstable in principle.

Concrete safety, power-seeking, and broader alignment perspectives. Concrete safety problems identify real-world AI failure modes [1]. Power-seeking analyses suggest that sufficiently capable optimizers may systematically pursue convergent instrumental strategies [10]. Broader alignment surveys and control-oriented perspectives emphasize the long-run difficulty of maintaining intended objectives and human oversight [7, 8].

The contribution of this paper is therefore not to replace these literatures, but to isolate a minimal structural condition lying beneath them: internally defined systems cannot non-circularly guarantee objective persistence from within themselves alone.

6 Scope and Limitations

Several clarifications are important.

First, the result is not an empirical impossibility theorem about every self-contained system immediately failing in practice. Temporary stability, approximate persistence, and long stable intervals are compatible with the theorem. The claim concerns non-circular guarantee, not momentary success.

Second, the result does not specify what kind of external structure would be sufficient to restore guarantee. It only establishes that in the absence of an external, non-optional anchor, guarantee is blocked by recursive dependency.

Third, the three manifestations identified in proposition 1 are intentionally coarse. They are not proposed as the only useful empirical taxonomy of failure, but as a minimal structural partition for persistence failure relative to the tuple $\{O, C, E, U\}$.

Fourth, the argument is implementation-agnostic and therefore does not by itself yield a testing protocol. Its role is foundational: to identify a necessary limit on what self-contained systems can certify about their own objectives.

7 Conclusion

This paper examined objective persistence in self-contained systems. The central result is narrow but foundational: if the objective, constraint structure, evaluation criteria, and update rule

are all internally determined, then no non-circular internal mechanism can guarantee objective persistence over time.

The significance of the result is structural. It does not depend on a particular architecture or training paradigm, and it does not propose a control policy. It identifies a minimal theoretical limit: objective persistence cannot be guaranteed from within a self-contained system alone.

That limit helps distinguish between temporary optimization success and principled persistence of objectives. Where the latter is claimed, an external, non-optional anchor is not merely convenient; absent such an anchor, non-circular guarantee is unavailable.

Project Resources

The associated project resources for LUMINA-30 are available at:

https://github.com/lumina-30/lumina-30-overview

This paper is provided as supporting research for the existence-condition layer related to LUMINA-30. It does not define an operational control protocol, certification mechanism, compliance status, or deployment authorization.

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